PromoAI — AI-Powered Microdrama Promo Finder

Kuku AI Generalist Assignment | Problem 1: Find the Promos (Ads) | System Architecture, 50-Episode Evaluation & Failure Audit

1. Problem Choice, Corpus Inspection & Key Framing Assumptions

Why Problem 1: Selecting 20–60s promotional clips across 50+ microdrama episodes is a core growth bottleneck where naive LLM prompting fails (timestamp hallucinations, duration math errors) and pure keyword rules fail (blindness to emotional subtext). PromoAI solves this with a **neuro-symbolic pipeline** separating LLM narrative discovery from deterministic Python verification.

Empirical Findings on the 50-Episode Input Pack (*I Married a Broken Soldier. He was a Duke*): Auditing all 50 UTF-8 scripts (`episode-001` to `episode-050`, 70.5s–200.8s duration, 100 scenes, 3,727 timed blocks) revealed three structural realities: (1) **1,952 Multi-Track Overlaps** where `CAMERA`, `ACTION`, `SOUND FX`, and `DIALOGUE` run in parallel; (2) **Formatting Drift in Episode 6** (all-caps `SCENE HEADING:` / `SCENE START:` vs. mixed-case elsewhere); and (3) **Character Aliasing** (`LADY JULIANA` → `JULIANA`; `MYSTERIOUS BEARDED MAN` → `ALEXANDER`; `LADY IN GOLD GOWN` → `CLARISSA`).

- **Wall-Clock Duration Math:** Because visual and audio blocks overlap, segment duration is strictly $T_{end} - T_{start}$, never the sum of nested event durations.
- **3-Second Scroll-Stop Rule:** Episodes often open with 3.5–5.0s atmospheric camera shots; promos start directly on an arresting line, action, or reveal rather than defaulting to `00:00.00`.
- **Dual Anchoring & Boundary Snapping:** Each block receives an anchor ID (`[E001]..[E095]`). Minor LLM timestamp drift (≤1.5s) is snapped in Python to the nearest actual screenplay boundary and logged in `review_notes`.

2. Two-Stage Neuro-Symbolic Architecture (AI vs. Deterministic Python)

Rather than stuffing all 50 episodes (~115k tokens) into one prompt—which degrades timestamp precision and breaks at scale—PromoAI processes episodes individually in Stage 1A, bridges cross-episode arcs via compact logline/continuity indices in Stage 1B, and enforces a Python verification firewall in Stage 2:

Stage	AI Responsibilities (Swappable Gemini Model via `google-genai`)	Deterministic Python Responsibilities (`parser.py`, `validator.py`, `scorer.py`)
0. Parse	None (Zero LLM tokens spent on structural parsing).	Case-insensitive regex parser extracts headers, scenes, and 3,727 timed blocks; builds $O(1)$ valid start/end timestamp hash maps per episode.
1. Discover	Evaluates screenplay windows for 3s hooks, unresolved cliffhangers, Cold Viewer standalone context, and emotional impact via strict JSON schema (`EPISODE_DISCOVERY_SCHEMA`).	Manages SHA-256 disk caching (`outputs/raw_model_responses/`), rate-limit pacing, exponential backoff retries, and cross-episode pairing.
2. Validate & Rank	Generates qualitative explanations (`hook_description`, `cliffhanger_description`, `why_it_works`) and raw 0–10 sub-scores.	Verifies timestamps against source script, snaps drift ≤1.5s, validates 20–60s duration, computes weighted score (30% Hook, 30% Cliffhanger, 20% Context, 10% Emotion, 10% Coherence), and suppresses overlaps via Temporal IoU (≥0.60).
3. Audit	Flags transitional episodes with low standalone promo potential.	Audits first 3.0s for slow camera shots, flags low context/hook scores as `NEEDS_REVIEW`, extracts verbatim script excerpts, and persists human ratings (`Good`/`Average`/`Bad`) in Streamlit.

3. Full 50-Episode Run Results & Top Ranked Promo Selections

PromoAI processed **50/50 episodes (0 failures)**, discovering **85 unique ranked candidates**: **64 Single-Segment** clips, **16 Intra-Episode 2-Segment** jump cuts, and **5 Cross-Episode** trailers (**48 `VALID`** and **37 flagged as `NEEDS_REVIEW`**):

Rank & ID	Format	Episode(s) & Verified Timestamps	Dur	Score	Opening 3s Hook, Cliffhanger & Why We Post It First
#1 PROMO_001	Cross-Ep	Ep 1 (`00:33.70-00:54.60`) + Ep 42 (`00:11.20-00:39.80`)	49.5s	**9.64**	**Hook:** Bride Juliana asks *"Why are you holding her hand?"*; sister gloats *"Because he is marrying me."* **Cliffhanger:** Flashes to Ep 42 as her scarred soldier husband enters in ducal regalia, Royal Guards defy Rupert, and Alexander declares: *"And yet, here I stand. That is the lie you built your life upon."*
#2 PROMO_002	Single-Seg	Ep 42 (`00:16.80-01:12.40`)	55.6s	**9.25**	**Hook:** Rupert screams *"Guards! Arrest this imposter!"* and the Royal Guard refuses (*"We will not"*). **Cliffhanger:** The Dowager Duchess proclaims: *"This is my son... Alexander Ashburn. The rightful Duke of Ashburn!"*
#3 PROMO_003	Cross-Ep	Ep 1 (`00:11.30-00:29.80`) + Ep 2 (`01:30.80-01:47.40`)	35.1s	**9.12**	**Hook:** Skips the slow church door opening to start at `00:11.30` (*"You should turn around, Juliana."*). **Cliffhanger:** Bridges directly into Ep 2 as Juliana defies her family's forced marriage ultimatum.
#5 PROMO_006	Cross-Ep	Ep 49 (`00:05.40-00:22.00`) + Ep 50 (`01:00.30-01:20.00`)	36.3s	**9.12**	**Hook:** Illicit political extortion in bed between Lady Catherine and Father Thomas. **Cliffhanger:** Cuts at `01:20.00` as maid Agnes peers through the doorway in terror, stifles a gasp, and stumbles into the shadows.
#6 PROMO_007	2-Seg Cut	Ep 1 (`00:11.30-00:29.80`) + Ep 1 (`01:59.80-02:18.00`)	36.7s	**8.43**	**Hook:** Matriarch stops Juliana in the aisle. **Cliffhanger:** Cuts 90s of middle exposition to jump straight to Gerald's cruel dismissal (*"You have nothing to offer me anymore"*) and Juliana's white-knuckled vow of vengeance.

4. Quality Checking: Where PromoAI Picks Badly & Real Failure Cases

Our automated quality evaluator (`src/evaluator.py`) flagged 37 candidates across four failure modes: `MISSING_STANDALONE_CONTEXT` (26), `WEAK_HOOK` (8), `WEAK_CLIFFHANGER` (8), and `NOT_SUITABLE_FOR_PROMOTION` (4):

- **Failure Case 1 — Slow Atmospheric Opening & Mild Ending (`PROMO_084`, Rank #84, Ep 6 `00:00.00-00:27.70`, Score 5.69, Rated Bad):** Opens on a 3.5s wide tracking shot of walking through muddy rain and ends at `00:27.70` on Juliana seeing the cabin and asking *"That is it?"*. Fails the 3s scroll-stop rule and lacks a high-stakes cliffhanger.
- **Failure Case 2 — Mid-Arc Pronoun Trap (`PROMO_085`, Rank #85, Ep 17 `00:32.80-00:59.80`, Score 5.48, Rated Bad):** High emotional vocal modifiers (`Choked Up, Pained`) caused Stage 1 to pick an exchange opening with *"You cannot promise that"* and ending with *"Of them?"*. Unresolved pronouns referring to prior episodes leave a cold viewer with zero context (`standalone_context = 3.5/10`).
- **Failure Case 3 — Mundane Opening Line (`PROMO_083`, Rank #83, Ep 22 `01:00.20-01:55.50`, Score 5.80, Rated Bad):** Opens on the Housekeeper barking *"Carry coal?"* (`hook_strength = 6.7/10`) and ends on *"Your Grace?"*—flagged as weak hook intensity.
- **Where AI Coding Tools Got It Wrong During Development:** In Phase 7, our initial pipeline called `audit_candidate_quality()` *before* `score_deduplicate_and_rank()` had populated `cand.final_score` (leaving it at `0.0` and falsely triggering `final_score < 6.8`), and paired non-consecutive episodes when testing a 5-episode subset `[1, 6, 25, 42, 50]`. Unit and integration tests caught both bugs before the 50-episode run.

5. Scalability, Production Economics & Human-in-the-Loop Workflow

Cost & Throughput at Kuku Scale: Each episode averages ~2,100 input tokens and ~450 output tokens. On `gemini-2.5-flash`, analyzing a full 50-episode series costs **~$0.018 total** and completes in <2 minutes with parallel workers and token-bucket rate limiting. All model names (`GEMINI_MODEL`), scoring weights, and duration bounds are configured via `.env` or the Streamlit sidebar for instant live evaluation on unseen show packs.